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Vinay Ram Gazula

Graduate Student at NJIT

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Aspiring data scietist with a solid foundation in mechanical engineering, seeking a challenging role where I can leverage my analytical skills and engineering background to extract meaningful insights. Eager to contribute my expertise in machine learning, statistical analysis, and problem-solving to drive data-driven decision-making making in a dynamic and innovative environment.

Education

Master of Science, *New Jersey Institute of Technology* — **Sep 2023 - May 2025** | NJ, USA
Data Science Computational Track — GPA: 3.83
Relevant Coursework: Applied Statistics, Introduction to Big Data, Data Managment Systems Design, Advanced Database Systems Design, Machine Learning, Data Analytics with R Program.

Bachelor of Technology, *SRM University AP* — **Aug 2017 - May 2021** | AP, India
Mechanical Engineering — GPA: 3.52
Awarded with merit scholarship of 50% tuition fee for the entire Bachelor's.

Technical Skills

Programming Languages	Python, R, Bash, C/C++, Java, MATLAB, Pig, Hadoop
Database Languages	SQL, MySQL, SQLite, BigQuery, Oracle, PL/SQL, MongoDB
Data Analytics	Spreadsheets, Pandas, Tidyverse, Matplotlib, Plotly, Tableau
Tools	**Git** GitHub **IDE / Text Editor** VS Code, RStudio, Sublime Text, KATE, Vim **Documentation** MS Office, LaTeX, Markdown **Operating Systems** Linux, MacOS, Windows, UNIX
Libraries	NumPy, SymPy, Tensorflow, Keras, yfinance, ggplot
CAD & CFD Softwares	Solidworks 2018, CATIA V5, Tecplot 360, Gmsh, ANSYS Fluent, AutoCAD

Experience

Graduate Research Assistant — **Oct 2023 - Present**
Under Dr. Jason T. L. Wang (Professor, Computer Science, NJIT) — Newark, USA

- Integrated Explainable AI (XAI) tools and leveraged interpretability into machine learning and deep learning tools for space weather research in collaboration with NSF / NASA (`https://nature.njit.edu/solardb/`).
- Incorporated XAI tools LIME, SHAP, PDP and ALE plots into a transformer-based framework "SolarFlareNet", used to forecast whether a $\geq M$-class flare occures within $24, 48$ & 72 hours using SHARP parameters.
- Submitted a conference paper in 37th International FLAIRS Conference which is currently under review.

CAD Designer — **Mar 2022 - Apr 2023**
Jayram Buildeng Solutions Pvt. Ltd. — Vijayawada, India

- Designing 2D CAD drawings using AutoCAD.
- Producing high quality, accurate, and timely reports for clients.
- Actively organised and participated in client meetings and contributed in promoting the services provided by the firm.

Research Intern — **Sep 2021 - Feb 2022**
Under Dr. Satya P. Jammy (Associate Professor, Mechanical Department, SRM University AP) — Amaravati, India

- Worked on a Science Education and Research Board (SERB) govt. of India project titled "Wall Effects in Shock Wave Boundary Layer Interaction".
- Contibuted in the development of solver code for gaseous reactions and viscous properties.
- Validated the solver for different flow configurations like Single ramp, Double ramp, etc,.
- Published a conference paper in 26th National and 4th International ISHMT-ASTFE Heat and Mass Transfer Conference.

Research Publications

FLAIRS-37 — **May 2024**
The Florida Artificial Intelligence Research Society (FLAIRS) — In Review

- Vinay, G., Yasser A., Jason T., "An Interpretable Transformer Model for Operational Flare Forecasting".

IHMTC-2021 — **Dec 2021**
ISHMT-ASTFE Heat and Mass Transfer Conference — Publication

- Vinay, G., Dinesh, D., Satya, P., "Accelerating a Hypersonic CO_2 Reaction Solver", ***Proceedings of the 26th National and 4th International ISHMT-ASTFE Heat and Mass Transfer Conference December 17-20, 2021, IIT Madras, Chennai-600036, Tamil Nadu, India.***

Professional Certifications

Google Data Analytics **Dec 2022 - Apr 2023**
Coursera Certificate

Projects

Interpretable Machine Learning for Flare Prediction | *Python, Tensorflow, ML, DL, XAI* **Oct 2023 - Ongoing**
Personal Project Project Source

- Integrated XAI tools like LIME, SHAP, PDP and ALE plots into multiple machine learning and deep learning tools used for predictiong space weather.
- Pioneered the integration of interpretability techniques into transformer-based machine learning model named "SolarFlareNet".
- Studied the feature importance and feature interactions between the SHARP magnetic parameters used to forecast solar flares for NOAA selected active regions.

eComputer Store Database System | *MySQL, Python, Streamlit, Pandas* **Sep 2023 - Dec 2023**
Data Managment Systems Design Course Project Project Source

- Designed and developed a working database for an eComputerStore with all mentioned functionalities.
- Followed a traditional approach for building database by designing an Enhanced Entity Relationship (EER) diagram, transforming the EER model to a Relational database schema and creating the database tables in MySQL using the Relational schema.
- Developed a web application using Streamlit which connects to the MySQL server using a pipeline.

MURPH Analytics | *Python, Streamlit, Pandas, Matplotlib, Plotly* **Mar 2023 - May 2023**
Personal Project Project Source

- Built an interactive dashboard using Streamlit to post-process and perform different analytical studies on the simulation data generated from the MURPH Solver.
- Implemented unstructured mesh plotting and 2D contour plots using Plotly.

MURPH | *Python, SymPy, NumPy, OpenMP, CUDA, OP2-API, CFD* **Dec 2022 - Ongoing**
In collaboration with Dr. Satya P. Jammy (Associate Professor, Mechanical Department, SRM University AP) Project Source

- A Multi-dimensional (2-D/3-D), Unstructured-mesh and Reactions based Parallel solver for Hyper-sonic flow regimes.
- New design uproach is used to build the solver code so that the parallel code generation can be fully automated.
- 2D Ideal gas solver code is developed and is validated using existing literature.
- Work is underway for integrating species reactions and development of a 3D solver.

AlgoTrade API | *Python, yfinance, Pandas, Tensorflow, ks-api-client* **May 2022 - Oct 2022**
Personal Project Project Source

- Fully automated NSE Stock/Equity trading-bot for Kotak Securities (KOTAK TRADE API) with integrated back testing and ML algorithms for price and trend predictions.
- The bot gathers any desired stock's historical price data, performs technical analysis using technical indicators and places orders to the stock broker using the broker API.

Accelerating a Hypersonic CO_2 Reaction Solver | *Python, SymPy, NumPy, OpenMP, CUDA, OP2-API, CFD* **Jul 2020 - May 2021**
B.Tech Final Year Project

- A 2-D unstructured finite volume method based solver capable of simulating hypersonic flows in CO_2 rich Martian atmosphere with chemical kinetics for eight species consisting of $N_2, O_2, NO, O, N, CO_2, CO$ and C incorporated in it is developed using pYthon.
- The solver is parallelized to run on single or multiple CPUs and Nvidia GPUs in order to attain greater computational speeds using OP2 application programming interface.

Certifications

Algorithmic Trading & Quantitative Analysis Using Python **Mar 2023 - Apr 2023**
Udemy *Certificate*

Introduction to Programming with MATLAB **Apr 2020 - May 2020**
Coursera *Certificate*

Autodesk Fusion 360 Integrated CAD/CAM/CAE **Feb 2020 - Mar 2020**
Coursera *Certificate*

Robo Feast 2018 **Feb 2018**
Certificate of Participation
Participated in the 2 days National Level Workshop cum Championship on "*WiFi-Controlled Robotics*" organized by ***Revert Technology Pvt. Ltd. & EDC, IIT Roorkee.***